

CENTIPEDE Implementation

This is an R implementation of the CENTIPEDE model first proposed by R Pique Regi et al. 2011, *Accurate inference of transcription factor binding from DNA sequence and chromatin accessibility data*. Though the model was originally implemented for DNase-seq datasets, we extend its applications to single nuclei ATAC-seq data.

Model notation – we have matrix $X \in \mathbb{R}^{N \times 200}$, where each row corresponds to the transcription factor binding site (TFBS) in question, and each column corresponds to the base pair (bp) position flanking the TFBS, from -100 (upstream) to 100 (downstream). The values of X indicate the number of cutsites detected at position j . We denote X_i as row i of X . We denote $R_i = \sum_j X_{ij}$. Additionally, we introduce a latent variable, Z_i , to denote whether TFBS i is bound ($Z_i = 1$) or not ($Z_i = 0$).

Marginalizing out Z , we have:

$$P(Data) = \pi P(Data | Z = 1) + (1 - \pi) P(Data | Z = 0)$$

Where $Data$ is our Tn5 cutsite data around each motif site, and $\pi \in [0, 1]$ is the mixture proportion. The Tn5 distribution around a motif is parameterized as a multinomial distribution conditional on the number of Tn5 cutsites observed within the window and the binding state (bound/unbound), and the total number of cutsites is parametrized as a negative binomial:

$$P(X_i | Z_i = 1, R_i, p, \mu_1, \phi_1) = Multinomial(X_i | p, R_i) NB(R_i | \mu_1, \phi_1)$$

$$P(X_i | Z_i = 0, R_i, \lambda, \mu_0, \phi_0) = Multinomial(X_i | \lambda, R_i) NB(R_i | \mu_0, \phi_0)$$

Where p is a vector of multinomial probabilities, $\lambda = [\frac{1}{200}, \dots, \frac{1}{200}]$ is a vector of equal probabilities parameterizing a uniform multinomial distribution, μ_1 and ϕ_1 are the mean and dispersion for the bound motif negative binomial, and μ_0 and ϕ_0 are the mean and dispersion of the unbound negative binomial.

The complete likelihood (if Z were known) is

$$L(X|Z, R) = \prod_i^N (\pi Multinomial(X_i | p, R_i) NB(R_i | \mu_1, \phi_1))^{1\{Z_i=1\}} \\ \cdot ((1 - \pi) Multinomial(X_i | \lambda, R_i) NB(R_i | \mu_0, \phi_0))^{1\{Z_i=0\}}$$

And the corresponding log likelihood

$$l(Z|X, R, \theta) = \sum_i^N 1\{Z_i = 1\} \log \pi + 1\{Z_i = 1\} X_i \log(p) + 1\{Z_i = 1\} (\log(\Gamma(R_i + \phi_1)) \\ - \log \Gamma(\phi_1) - \log \Gamma(R_i + 1) + \phi_1 \log(\phi_1) \\ - \phi_1 \log(\phi_1 + \mu_1) + R_i \log(\mu_1) - R_i \log(\phi_1 + \mu_1)) \\ + 1\{Z_i = 0\} \log(1 - \pi) + 1\{Z_i = 0\} X_i \log(\lambda) \\ + 1\{Z_i = 0\} (\log(\Gamma(R_i + \phi_0)) - \log \Gamma(\phi_0) \\ - \log \Gamma(R_i + 1) + \phi_0 \log(\phi_0) \\ - \phi_0 \log(\phi_0 + \mu_0) + R_i \log(\mu_0) - R_i \log(\phi_0 + \mu_0))$$

Where we denote $\theta = [\pi, p, \mu_1, \phi_1, \mu_0, \phi_0]$ for short. As Z_i is an unobservable latent variable, CENTIPEDE uses expectation maximization to estimate the parameters.

E-step: First, we calculate the expectation of Z given the initial parameter estimates

$$E(Z_i | \theta, X_i, R_i) = \frac{\pi \text{Multinomial}(X_i | p, R_i) \text{NB}(R_i | \mu_1, \phi_1)}{\pi \text{Multinomial}(X_i | p, R_i) \text{NB}(R_i | \mu_1, \phi_1) + (1 - \pi) \text{Multinomial}(X_i | \lambda, R_i) \text{NB}(R_i | \mu_0, \phi_0)}$$

M-step: replacing $1\{Z_i = k\}$ from the complete log likelihood with its expectation, the parameter p , representing the probabilities of the multinomial distribution, is updated as

$$p^{(t+1)} = \frac{\sum_j X_{ij} E(Z_i^{(t)} | \theta^{(t)})}{\sum_i \sum_j X_{ij} E(Z_i^{(t)} | \theta^{(t)})}$$

Where t is the current iteration.

Unfortunately, the parameters of the bound and unbound negative binomial distributions, μ_1, ϕ_1, μ_0 , and ϕ_0 have no closed form solutions. However, optimizing the associated log likelihood components is equivalent to a weighted generalized linear model, and are thus updated using the function `glm.nb` from the R package MASS, with the weights set to $E(Z_i | \theta^{(t)})$ for the bound case, and $1 - E(Z_i | \theta^{(t)})$ for the unbound case.

The quantity π is parametrized by a sigmoid function, and incorporates prior information:

$$\pi(\beta) = \frac{1}{1 + e^{-W\beta}}$$

Where W is a $m \geq 1$ column matrix containing the motif PWM scores and any additional prior information, such as the motif conservation scores or output from predictive machine learning models. β is optimized by the R function `optim`. We iterate the CENTIPEDE EM-algorithm until the negative log likelihood decreases by less than $1e-3$.

As TF-footprints have been known to have TF-specific signatures, we run the CENTIPEDE algorithm on each TF individually within each cell subtype. For downstream analysis, we select TFs with $P(\text{Bound} | \text{data}) = E(Z_i | \theta, X, R) \geq 0.95$.